HT_MK1 Harness Tester

Functional Test Procedure

Document ID	HT_MK1_FTP
Revision	1.1.0
Date	22-10-2026
Project	HT_MK1 Harness Tester - Infinity Integrations Pvt Ltd

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Approval of Document:

	Name	Designation	Date
Prepared By			22-10-2026
Reviewed By			22-10-2026
Approved By			22-10-2026

Table of Revision

Version	Note/ Comment	Date
1.0.0	Initial release	27-08-2026
1.1.0	Reissued in the standard document format; terminology aligned with the Mode 1 / Mode 2 GUI	22-10-2026

Distribution List (For limited distribution only)

S. No.	Name	Organization / Department
1		Infinity Integrations Pvt Ltd
2		
3		

Contact Details

S. No.	Name	Communication Details
1		
2		
Registered Office Address:		

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1 Purpose and Scope

1.1 Purpose

This procedure verifies that a built HT_MK1 unit performs its continuity, resistance, and insulation tests correctly, before the unit is accepted or shipped.

1.2 Scope

Covers functional (pass/fail) verification of one assembled unit: Control Card, Matrix Card, and HV Card(s), operated through the GUI. Does not cover firmware programming (see the Flashing Procedure) or day-to-day operation (see the User Manual). This procedure assumes both are already done and refers to them rather than repeating their steps.

1.3 Acceptance Criteria

The unit passes this procedure if every test case in Section 5 passes. Any failure is logged as a deviation (Section 7) and resolved before the unit is accepted.

2 References

- HT_MK1 User Manual: GUI operation, screen layout, safety.
- HT_MK1 Flashing Procedure: programming and firmware verification.
- HT_MK1 Code Walkthrough: architecture, protocol, and current known limitations.

3 Test Environment and Equipment

Item	Notes
Unit under test	Assembled Control Card + Matrix Card + HV Card(s), firmware programmed and verified per the Flashing Procedure.
GUI	Installed on a Windows PC, connected to the unit.
Reference harness (known-good)	A harness, or its netlist, with predetermined expected continuity, resistance, and insulation values.
Reference harness variants (faulted)	Harnesses or netlists with an intentional open, an intentional short, an out-of-limit resistance, and an intentional insulation fault, for the negative test cases below.
Reference resistor or wire of known resistance	Used to check the resistance measurement against a known value (Section 5.3).

Table 1: Equipment

4 Preconditions

- Firmware programmed and the PING/PONG check completed (Flashing Procedure, Section 6).
- GUI installed and able to connect to the unit.
- Operator has read the User Manual, Section 3 (Safety), before Section 5.5 of this procedure.

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5 Test Cases

5.1 TC-01: Power-On and Connection

Objective: Confirm the unit powers on, the firmware boots, and the GUI connects successfully.

Preconditions: None.

Steps:

1. Power on the unit.
2. Launch the GUI (User Manual, Section 4.1).
3. Connect to the instrument (User Manual, Section 4.2).

Expected Result: The status bar shows the instrument connected, state idle, and no faults.

Result	Actual / Measured Value	Pass / Fail	Tested By	Date

Table 2: Result record, TC-01

5.2 TC-02: Continuity Test (Mode 1)

Objective: Confirm the continuity test correctly passes a known-good harness and correctly flags an intentional open and an intentional short.

Preconditions: Harness on the Matrix connector. Reference harness variants available (Section 3).

Steps:

1. Load the known-good reference netlist (User Manual, Section 5).
2. Run continuity in Netlist mode (User Manual, Section 6.1). Record the result.
3. Repeat using the intentional-open variant. Record the result.
4. Repeat using the intentional-short variant. Record the result.

Expected Result: Known-good harness: all nets pass. Intentional open: the affected net is reported missing (fault F06). Intentional short: the affected net is reported unexpected (fault F07).

Result	Actual / Measured Value	Pass / Fail	Tested By	Date

Table 3: Result record, TC-02

5.3 TC-03: Resistance Test (Mode 1)

Objective: Confirm the resistance test reports an accurate measurement for a wire of known resistance, and correctly flags an out-of-limit wire.

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Preconditions: TC-02 passed on the known-good harness.

Steps:

1. Run resistance on the known-good harness (User Manual, Section 6.1).
2. Record the measured value for the reference wire of known resistance.
3. Repeat using the out-of-limit resistance variant. Record the result.

Expected Result: The known-resistance reference measures within the accuracy stated in the Code Walkthrough (reference-resistor ratiometric method) of its known value. The out-of-limit wire is reported fail_high.

NOTE: This is the first real-hardware check of the ratiometric resistance measurement noted as unverified in the Code Walkthrough's Known Limitations. Record the measured value for the project record regardless of pass/fail.

Result	Actual / Measured Value	Pass / Fail	Tested By	Date

Table 4: Result record, TC-03

5.4 TC-04: Fixture Handover

Objective: Confirm HV testing is gated behind a passing Mode 1 and an explicit fixture move.

Preconditions: TC-02 and TC-03 both passed.

Steps:

1. Confirm the Run screen prompts to move the harness to the HV connector (User Manual, Section 6.2).
2. Attempt to run the insulation test before confirming the move. The GUI asks whether to continue without the Mode 1 tests; answer No.
3. Move the harness and confirm the move in the GUI.

Expected Result: The insulation test is refused/locked until the move is confirmed in the GUI, and unlocks immediately once confirmed.

Result	Actual / Measured Value	Pass / Fail	Tested By	Date

Table 5: Result record, TC-04

5.5 TC-05: Insulation Test (Mode 2)

Objective: Confirm the insulation test arms, applies HV, measures leakage, and returns the system to a safe state, for both a passing and a failing net.

Preconditions: TC-04 completed. Harness on the HV connector. Operator has read User Manual Section 3.

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Steps:

1. Run insulation on the known-good harness (the GUI arms the instrument as part of the run) (User Manual, Section 6.2).
2. Confirm the HV pill in the status bar shows "HV Safe" after the run.
3. Repeat using the intentional insulation-fault variant.

Expected Result: Known-good harness: all nets pass, the HV rail returns to 0 V, and "Safe" is shown. Fault variant: the affected net fails, and the system still returns to a safe state.

NOTE: This is the first real-hardware check of the insulation/HV path noted as unverified in the Code Walkthrough's Known Limitations.

Result	Actual / Measured Value	Pass / Fail	Tested By	Date

Table 6: Result record, TC-05

5.6 TC-06: Abort

Objective: Confirm Abort stops a running test immediately and returns the system to a safe state.

Preconditions: A harness with enough nets to observe a run in progress.

Steps:

1. Start a continuity or resistance run.
2. Press the round Stop button partway through the run (User Manual, Section 3.3). The log shows `ABORT sent`.

Expected Result: The run stops immediately, no further results stream in, and the status bar shows idle and "HV Safe".

Result	Actual / Measured Value	Pass / Fail	Tested By	Date

Table 7: Result record, TC-06

5.7 TC-07: Fault Handling

Objective: Confirm a fault is reported clearly and can be cleared.

Preconditions: None.

Steps:

1. Induce a fault condition (for example, a card intentionally unseated or disconnected).
2. Confirm the fault appears in the Faults panel with a description.
3. Resolve the condition and clear the fault (User Manual, Section 8).

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Expected Result: The fault is displayed with a description while active; clearing it returns the system to idle.

Result	Actual / Measured Value	Pass / Fail	Tested By	Date

Table 8: Result record, TC-07

5.8 TC-08: Results Export

Objective: Confirm a completed run's results can be exported.

Preconditions: At least one completed run from the test cases above.

Steps:

1. Go to the RESULTS screen (User Manual, Section 7).
2. Export a CSV for a completed run.

Expected Result: A CSV file is produced containing that run's results.

Result	Actual / Measured Value	Pass / Fail	Tested By	Date

Table 9: Result record, TC-08

6 Test Completion Summary

Test ID	Description	Pass / Fail
TC-01	Power-On and Connection	
TC-02	Continuity Test	
TC-03	Resistance Test	
TC-04	Fixture Handover	
TC-05	Insulation Test	
TC-06	Abort	
TC-07	Fault Handling	
TC-08	Results Export	

Table 10: Test completion summary

7 Deviations and Overall Verdict

List any test case that did not pass as written, what was observed, and how it was resolved. Leave blank if none.

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Test ID	Deviation Observed	Resolution

Table 11: Deviations

Overall Verdict: ☐ PASS ☐ FAIL

Any item still open at the project level (not specific to this unit) is tracked in the Code Walkthrough's Known Limitations section, not here.

8 Sign-off

Field	Value
Unit serial number	
Firmware version tested	
Tested by	
Date	
Approved by	
Date	

Table 12: Sign-off record

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